

Riccardo Orlando

PH.D. STUDENT, 3RD YEAR · DEEP LEARNING AND NATURAL LANGUAGE PROCESSING

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Experience

Researcher

Rome, Italy

SAPIENZA UNIVERSITY OF ROME

Nov. 2021 - Present

- Trained the first family of Open Foundation Models for the Italian language (Minerva (1B, 3B, and 7B) LLMs).
- Developed a novel retrieval-augmented system that achieves state-of-the-art performance in Entity Linking and Relation Extraction.
- Led the data annotation process that resulted in a novel dataset for Semantic Role Labeling.
- Developed production-ready pipelines for several state-of-the-art NLP tasks, including Word Sense Disambiguation, Semantic Role Labeling, and Semantic Parsing.
- Served as a Teaching Assistant for Natural Language Processing and Elective in Artificial Intelligence MSc courses.

NLP Engineer

Rome, Italy

BABELSCAPE

Feb. 2021 - Dec. 2021

- Developed end-to-end state-of-the-art multilingual NLP pipelines.
- Implemented a Machine Learning and NLP pipeline serving framework, drastically reducing deployment time.
- Authored and presented two papers at EMNLP 2021.

Software Engineer

Rome, Italy

CAPGEMINI

Mar. 2018 - Sept. 2018

- Developed a back-end application using Spring Boot to handle internal workflows for different customers.
- Contributed to the front-end application, which was built on top of Angular.

Education

Ph.D. in Computer Science (Deep Learning & Natural Language Processing)

Rome, Italy

SAPIENZA UNIVERSITY OF ROME

Nov. 2021 - Present

- Advisor: Roberto Navigli

Master of Science in Engineering in Computer Science

Rome, Italy

SAPIENZA UNIVERSITY OF ROME

Oct. 2018 - Jan. 2021

- Final Grade: 110/110 with honors
- Thesis: Automatic Approaches to Produce Multilingual Training Data for Semantic Role Labelling.

Bachelor of Science in Computer Science

Rome, Italy

SAPIENZA UNIVERSITY OF ROME

Oct. 2014 - May 2018

- Final Grade: 101/110
- Thesis: Densest Subgraph in Fork/Join.

Relevant Publications

2024	Francesco Maria Molfese, Simone Conia, <u>Riccardo Orlando</u> , and Roberto Navigli. ZEBRA: Zero-Shot Example-Based Retrieval Augmentation for Commonsense Question Answering.	EMNLP 2024
2024	<u>Riccardo Orlando</u> , Luca Moroni, Pere-Lluís Huguet Cabot, Edoardo Barba, Simone Conia, Sergio Orlandini, Giuseppe Fiameni, and Roberto Navigli. Minerva LLMs: The First Family of Large Language Models Trained from Scratch on Italian Data.	arXiv
2024	<u>Riccardo Orlando</u> , Pere-Lluís Huguet Cabot, Edoardo Barba, and Roberto Navigli. ReLiK: Retrieve and Link, Fast and Accurate Entity Linking and Relation Extraction on an Academic Budget.	ACL 2024
2024	Simone Conia, Edoardo Barba, Abelardo Carlos Martínez Lorenzo, Pere-Lluís Huguet Cabot, <u>Ricardo Orlando</u> , Luigi Procopio, Roberto Navigli. MOSAICo: a Multilingual Open-text Semantically Annotated Interlinked Corpus.	NAACL 2024
2023	<u>Riccardo Orlando</u> , Simone Conia, and Roberto Navigli. Exploring Non-Verbal Predicates in Semantic Role Labeling: Challenges and Opportunities.	ACL 2023
2022	<u>Riccardo Orlando</u> , Simone Conia, Stefano Faralli, and Roberto Navigli. Universal Semantic Annotator: the First Unified API for WSD, SRL and Semantic Parsing.	LREC 2022

2021	Riccardo Orlando , Simone Conia, Fabrizio Brignone, Francesco Cecconi, and Roberto Navigli. AMuSE-WSD: An All-in-one Multilingual System for Easy Word Sense Disambiguation.	EMNLP 2021
2021	Simone Conia, Riccardo Orlando , Fabrizio Brignone, Francesco Cecconi, and Roberto Navigli. InVeRo-XL: Making Cross-Lingual Semantic Role Labeling Accessible with Intelligible Verbs and Roles.	EMNLP 2021

Skills

Frameworks	Deep Learning & Machine Learning (PyTorch, PyTorch Lightning, Hydra, Transformers, ONNX, TensorFlow, scikit-learn)
Back-end	FastAPI, Django, REST API
Languages	Python, Java, Kotlin, Bash/Shell
Developer Tools	Git, Docker, VS Code, Visual Studio, PyCharm, IntelliJ

Honors & Awards

2024	Winner , 3x Italian SuperComputing Resource Allocation (ISCRA) Call for Proposal type C for compute hours at CINECA, the Italian HPC Center	Rome, Italy
2023	Winner , Italian SuperComputing Resource Allocation (ISCRA) Call for Proposal type B for compute hours at CINECA, the Italian HPC Center	Rome, Italy
2022	Winner , Call for Proposals for Scientific Research	Rome, Italy

Program Committees

2024	Reviewer , ARR February 2024, ARR June 2024
2023	Reviewer , ARR December 2023, ARR October 2023, EMNLP 2023, ACL 2023, EACL 2023
2022	Reviewer , LREC 2022
2021	Reviewer , EMNLP 2021